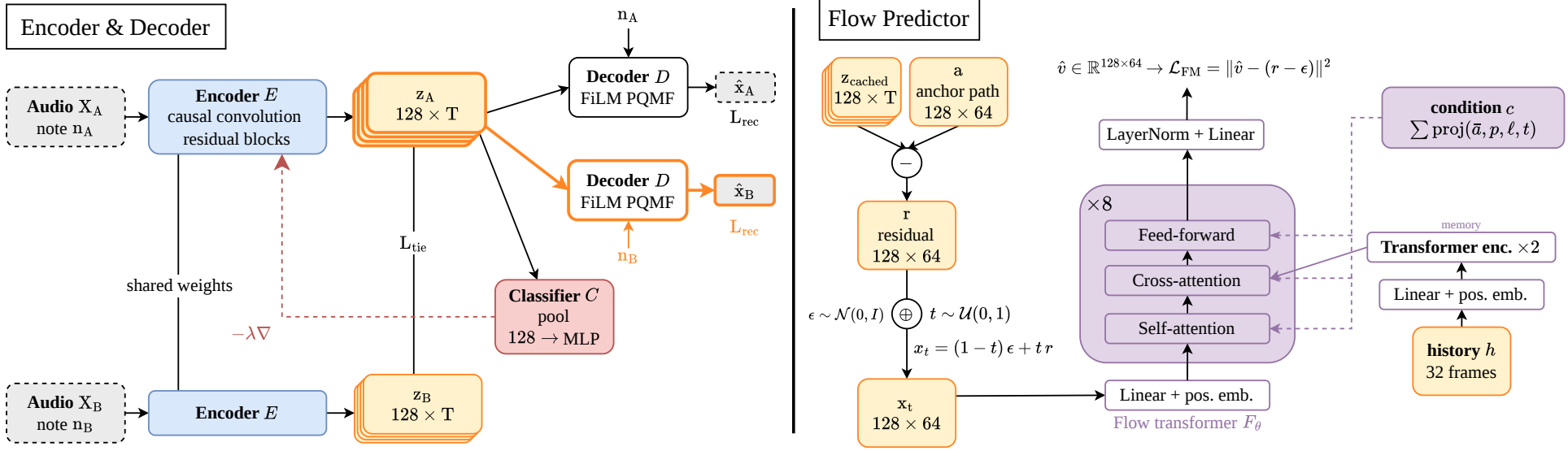

Encoder & Decoder
Audio X_A
note n_A
Encoder E
causal convolution
residual blocks
z_A
128 × T
n_A
Decoder D
FiLM PQMF
$\hat{x}_A$
L_{rec}
Decoder D
FiLM PQMF
$\hat{x}_B$
L_{rec}
n_B
L_{tie}
shared weights
$-\lambda\nabla$
Classifier C
pool
128 → MLP
Audio X_B
note n_B
Encoder E
z_B
128 × T
Flow Predictor
z_{cached}
128 × T
a
anchor path
128 × 64
r
residual
128 × 64
$\epsilon \sim \mathcal{N}(0, I)$
$t \sim \mathcal{U}(0, 1)$
$x_t = (1-t)\,\epsilon + t\,r$
x_t
128 × 64
$\hat{v} \in \mathbb{R}^{128\times64} \to \mathcal{L}_{FM} = \|\hat{v} - (r - \epsilon)\|^2$
LayerNorm + Linear
×8
Feed-forward
Cross-attention
Self-attention
Linear + pos. emb.
Flow transformer F_θ
condition c
$\sum \text{proj}(\bar{a}, p, \ell, t)$
memory
Transformer enc. ×2
Linear + pos. emb.
history h
32 frames